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Impact of AI-Generated Phishing Attacks: A New Cybersecurity Threat

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Abstract. As generative artificial intelligence (AI) technologies, such as OpenAI's GPT-4 and other large language models (LLMs), advance, the cybersecurity landscape faces a pressing challenge: the rise of AI-generated phishing attacks. These attacks exploit AI's ability to convincingly mimic human language at scale, making them significantly harder to detect than traditional, human-generated phishing attempts. The purpose of this study is to address this growing threat by analysing the unique characteristics of AI-driven phishing emails, focusing on the linguistic and contextual features that distinguish them from human-authored emails. The research identifies a critical limitation in current phishing detection systems, which exhibit low precision and recall when dealing with AI-generated content. To solve this issue, we propose a novel detection framework that combines advanced machine learning and natural language processing (NLP) techniques with behavioural analysis. Experimental results demonstrate the effectiveness of the proposed model, achieving a precision rate of 91% and a recall rate of 89% and an F1 score of 90% in identifying AI-generated phishing emails—substantially outperforming existing systems. This paper's main contributions include a comprehensive analysis of AI-generated phishing email characteristics, the development of an innovative detection framework, and the presentation of adaptive strategies to enhance cybersecurity defences against the growing threat of generative AI.

Keywords— AI-generated phishing, generative AI, large language models (LLMs), cybersecurity, phishing detection, natural language processing (NLP), email security, machine learning, AI-driven attacks.

1 Introduction

Phishing remains one of the most widespread and damaging cybersecurity threats, with its prevalence continuing to rise in an increasingly digital world. According to the Anti-Phishing Working Group (APWG), over 1.3 million phishing attacks were reported in 2024, representing a 15% increase from the previous year and accounting for a significant proportion of data breaches [1]. Traditionally, phishing attacks have relied on social engineering techniques that exploit human vulnerabilities. These human-generated phishing emails often have noticeable flaws, like grammatical errors, suspicious URLs, or inconsistencies. These issues make it easier for detection systems—like content filters and domain blacklists—to reduce their impact.

However, recent advances in artificial intelligence (AI), particularly the development of generative AI models like OpenAI's GPT-4 and other large language models (LLMs), have transformed the landscape of phishing attacks. These models enable the creation of highly sophisticated and contextually tailored phishing emails that are nearly indistinguishable from legitimate communications. Unlike traditional phishing emails, AI-generated phishing emails lack the obvious errors that detection systems typically rely on, posing a significant challenge to existing cybersecurity measures [2]. This evolution has facilitated the emergence of a new class of attacks, including highly targeted campaigns such as business email compromise (BEC) and spear-phishing, executed with unprecedented scale and effectiveness.

Despite the critical importance of addressing this threat, existing phishing detection frameworks remain inadequate for combating AI-generated phishing emails. Current systems, which primarily use heuristic analysis,

rule-based filtering, or traditional machine learning approaches, struggle to detect the detailed and sophisticated features of AI-generated content. Furthermore, the rapid adoption of generative AI technology introduces challenges such as evolving attack strategies and the lack of comprehensive datasets to train detection systems. These gaps underscore the urgent need for innovative and adaptive approaches to detect and mitigate this emerging threat.

This study addresses these challenges by exploring the unique linguistic and contextual characteristics of AI-generated phishing emails and proposing a novel detection framework that integrates advanced machine learning, natural language processing (NLP), and behavioural analysis. The importance of this research lies in its ability to bridge a critical gap in current cybersecurity defences while advancing our understanding of how generative AI is reshaping the threat landscape. By achieving a precision rate of 91%, a recall rate of 89%, and an F1 score of 90% in experimental evaluations, the proposed framework demonstrates significant improvements over existing systems. This contribution is not only timely but also essential for equipping organisations with the tools needed to combat the evolving threat of AI-driven phishing attacks. The following sections provide valuable insights into the research conducted on AI-generated phishing attacks and the proposed detection framework:

Section 2: Related Work
This section reviews the evolution of phishing detection, from traditional signature-based and heuristic methods to advanced machine learning and NLP approaches. While these systems effectively detect human-generated phishing emails, they struggle against AI-generated emails, which are highly realistic and target-specific. Studies highlight gaps in current defences, emphasizing the need for improved models to address AI-driven phishing threats [3, 4, 6–10].

Section 3: Research Gaps/Research Problems
The paper identifies three critical research gaps: (1) the inability of current systems to detect the linguistic and contextual differences in AI-generated phishing emails, which are often fluent and contextually tailored, (2) the lack of generalised detection mechanisms that can adapt to new AI-driven phishing tactics, and (3) the limitations of traditional machine learning models, which are often unable to analyse the underlying content and structure of emails effectively. These gaps point to the need for more adaptive and content-aware detection frameworks.

Section 4: Proposed Solution/Methodology
In this section, the paper outlines the proposed methodology for detecting AI-generated phishing emails. The approach combines linguistic feature extraction, behavioural pattern analysis, and contextual evaluation. By leveraging machine learning and natural language processing (NLP), the framework offers a more comprehensive analysis of both the email content and user behaviour, which enhances its ability to detect phishing attempts compared to traditional systems.

Section 5: Results
The experimental results presented in this section demonstrate that the proposed framework achieves a precision rate of 91% and a recall rate of 89%, significantly outperforming existing phishing detection systems. The comparative analysis of AI-generated and human-generated phishing emails further illustrates the framework's effectiveness in distinguishing between the two types of attacks, underscoring the relevance of incorporating linguistic and contextual features into phishing detection.

Section 6: Conclusion
The conclusion explains the emerging challenges posed by generative AI technologies in detecting AI-generated phishing emails. It highlights the limitations of traditional detection systems and presents a hybrid machine learning and natural language processing framework that significantly improves detection accuracy. The proposed model achieved a precision rate of 91%, recall rate of 89%, and an F1 score of 90% in distinguishing AI-generated phishing emails from legitimate messages. The study emphasizes the need for adaptive, multi-layered defences that combine linguistic, behavioural, and contextual analysis to effectively counter evolving AI-driven threats. The research underscores the importance of updating detection models to stay ahead of AI advancements in phishing attacks.

Section 7: Future Work
Several potential areas for future improvement are highlighted in this section. These include the implementation of real-time analysis of user interactions for more immediate detection of phishing attempts, the integration of

blockchain-based authentication to prevent fraudulent emails from reaching inboxes, and the expansion of detection capabilities to cover multi-platform environments, such as SMS and social media. Additionally, the development of adaptive learning models that can continuously update in response to evolving phishing strategies is emphasized as a critical direction for future research.

2 Related Work

Phishing detection has seen significant advancements, transitioning from basic signature-based techniques to more sophisticated machine learning and deep learning models that analyse content, metadata, and user behaviour to identify phishing attempts [3]. Traditional phishing detection methods often rely on:

- Keyword-based filtering, where suspicious URLs or common phishing phrases are flagged.
- Heuristic approaches that analyse anomalies in email headers, sender information, and metadata.
- Machine learning classifiers, trained on large datasets of phishing and benign emails, which evaluate features like email structure, language patterns, and sender authenticity [4].

The emergence of generative AI models such as GPT-4, however, presents a unique challenge. These models can create highly realistic, grammatically correct, and personalised phishing emails that are far more difficult to detect using conventional methods. Unlike human-generated phishing emails, which often contain detectable errors, AI-generated emails can bypass keyword-based filters and conventional security mechanisms [5].

Do et al. [6] reported that existing phishing filters, especially keyword-based ones, are often ineffective against sophisticated phishing emails generated by AI. These AI-generated phishing attacks are particularly dangerous because they can be customised to specific targets, making them much more likely to succeed. In addition, the study investigated deep learning models for phishing detection, revealing that while these models improve accuracy, they are predominantly trained on datasets containing human-generated phishing emails. As a result, there is a gap in their ability to detect more advanced AI-generated phishing emails. [6].

Brown & Mitta et al. conducted an experiment on phishing emails crafted using GPT-3 and found that these AI-generated emails convincingly mimicked internal corporate communication, which significantly reduced the efficacy of existing detection systems. These findings underscore the growing gap in current defences against this type of attack [7, 8]. While Natural Language Processing (NLP)-based approaches have been extensively studied for phishing detection and in addition, Anderson et al. [9] proposed a model that analyses linguistic features such as sentence structure and word usage patterns, demonstrating improved performance in detecting phishing emails. However, these approaches are still vulnerable to the highly fluent, human-like language produced by generative AI models.

Furthermore, Mishra et al [10] explored detection methods targeting AI-generated phishing, focusing on identifying linguistic features unique to AI-generated content. Their findings revealed that such phishing emails often lack emotional depth or exhibit complex, yet unnatural, sentence structures, suggesting potential avenues for improving detection systems [10].

While these studies provide valuable insights, most existing research does not specifically address the increasing threat of AI-generated phishing emails [38]. This paper bridges this gap by conducting a comprehensive analysis of these AI-generated threats and proposing a novel detection model that integrates NLP and machine learning to enhance phishing detection in this new threat landscape.

3 Problem Statement / Research Gaps

The advancement of generative AI technologies has enabled attackers to craft sophisticated phishing emails that evade traditional detection mechanisms. Existing phishing detection filters, even those enhanced with machine learning (ML) models, are predominantly trained on human-generated phishing emails, which often contain obvious signs of deception, such as grammatical errors, suspicious URLs, and atypical formatting. By contrast, AI-generated phishing emails tend to eliminate these red flags, making them harder to identify through standard detection methods [11].

The primary research gaps addressed in this paper include:

1. Linguistic and Contextual Differences: Current phishing detection systems are limited in identifying the distinct linguistic patterns of AI-generated emails, which are often fluent, grammatically accurate, and contextually tailored to the recipient. Studies reveal that most traditional models fail to distinguish human-generated phishing from AI-generated phishing based on subtle language attributes, such as sentence structure, tone, and language complexity [12].

2. Lack of Generalised Detection Mechanisms: Existing phishing detection mechanisms are largely static and lack the adaptability required to detect new forms of AI-enabled phishing attacks. As noted by Nishitha [13], generative models like GPT-4 can adjust to diverse phishing contexts and scenarios, rendering many traditional, rule-based detection systems ineffective against these evolving threats [13].

3. Limitations of Current Machine Learning Approaches: While machine learning models are frequently used in phishing detection, these models rely heavily on predefined features such as email structure, sender reputation, and link patterns. They are generally ill-suited for detecting AI-generated phishing attacks, which possess high levels of linguistic sophistication and authenticity, closely resembling legitimate communication. This limitation underscores the need for detection models that are not only feature-based but also capable of analysing the underlying content and structure of emails in depth [14].

Thus, the central research problem addressed in this paper is the development of adaptive phishing detection models that can effectively recognise both traditional and AI-generated phishing emails. By identifying the specific characteristics of AI-generated phishing, the proposed models aim to improve detection capabilities and provide a more robust defence against evolving phishing threats.

4 Proposed Solution / Methodology

The proposed solution develops a robust detection system targeting AI-generated phishing emails through an integration of machine learning (ML), natural language processing (NLP), and behavioural analysis. This multi-layered approach addresses the unique challenges posed by AI-generated phishing by leveraging data collection, linguistic feature extraction, behavioural pattern analysis, and model training and evaluation as outlined below.

4.1 Data Collection

Accurate detection of AI-generated phishing requires comprehensive datasets to capture the distinct characteristics of both AI- and human-generated phishing emails.

- AI-Generated Phishing Dataset: A simulated dataset of AI-generated phishing emails is created using large language models (LLMs) like GPT-4 to generate phishing messages across various scenarios. These scenarios include impersonating executive communications, sending false invoices, and creating fake customer inquiries—common types of phishing attacks. By generating a wide range of AI-crafted phishing emails, this dataset will offer insights into the unique linguistic patterns and styles of AI, helping to effectively distinguish them from human-generated content [15].

- Human-Generated Phishing Dataset: A dataset of real-world phishing emails is compiled from public sources such as Kaggle [16] and the Email Phishing Validation Dataset [17], providing verified examples of human-generated phishing attempts. These emails often display common signs of deception, such as grammatical errors, unusual phrasing, and inconsistent formatting. This dataset will allow the detection model to compare and contrast these flaws with the typically fluent, contextually appropriate language produced by AI.

4.2 Linguistic Feature Extraction

Linguistic feature extraction forms a core component of this methodology, enabling distinctions between AI-generated and human-generated phishing emails. Using advanced NLP techniques, the following features will be extracted:

- Sentence Structure: AI-generated emails typically exhibit complex, well-structured sentences compared to human-generated phishing emails, which often contain grammatical inconsistencies or unusual phrasing. This structural disparity is due to the sophisticated training that LLMs undergo, resulting in fluently constructed sentences that resemble legitimate communication [18].
- Tone Consistency: AI-generated emails generally maintain a consistent tone and formal register, distinguishing them from the variable tones seen in human-generated phishing emails. AI-crafted messages that impersonate corporate communication may retain a professional tone, while human-generated emails frequently fluctuate between formal and informal language [19].
- Lexical Diversity: AI-generated emails often exhibit higher lexical diversity due to the vast range of vocabulary from diverse sources in their training datasets. In contrast, human-generated phishing emails may recycle specific phrases and keywords, resulting in lower lexical diversity. Analysing these patterns allows the detection model to use lexical diversity as a distinguishing factor [20].

4.3 Behavioural Pattern Analysis

In addition to linguistic features, behavioural patterns commonly seen in phishing attempts will be analysed to detect suspicious behaviours characteristic of phishing emails. This step is essential for identifying potential phishing intents, even when AI-generated language resembles legitimate communication.

- Response Urgency: Many phishing emails aim to provoke an immediate response. Behavioural analysis will identify phrases that convey urgency, such as “urgent,” “respond now,” or “account suspension,” which frequently appear in phishing emails. While traditional phishing detection also uses this technique, this approach will refine urgency detection to flag subtle prompts embedded within sophisticated AI-crafted messages [21].
- Content Personalisation: Phishing emails often include recipient-specific details, especially in spear-phishing attacks. AI-generated emails might feature these personalised details more consistently than traditional phishing attempts. Therefore, NLP models will be used to identify unusually high levels of personalisation, helping differentiate legitimate communication from phishing attacks [22].

4.4 Model Training and Evaluation

The detection model will be trained and evaluated using a hybrid approach, combining linguistic and behavioural indicators. This methodology consists of the following steps:

- Feature Engineering: A feature set will be constructed from the extracted linguistic and behavioural features, enabling the model to detect complex patterns characteristic of phishing emails. These features will include sentence complexity metrics, tone consistency scores, lexical diversity measures, and behavioural cues like urgency and personalisation [23].
- Model Selection: A hybrid machine learning model will be constructed, incorporating deep learning architectures such as Bidirectional Encoder Representations from Transformers (BERT) for NLP analysis and random forest classifiers for behavioural patterns. BERT's attention mechanism enables the model to capture subtle linguistic details, while the random forest model strengthens the analysis of behavioural patterns, resulting in a robust, multi-dimensional phishing detection system [24].
- Training and Validation: The hybrid model will undergo training with a stratified cross-validation approach on the AI- and human-generated datasets to ensure balanced performance. Evaluation metrics including accuracy, precision, recall, and F1-score will be used to measure the model’s effectiveness in detecting AI-generated phishing emails. Comparative benchmarks with existing phishing detection models will provide insight into the model's advancements in identifying sophisticated AI-based phishing attempts [25].
- Continuous Learning Mechanism: Recognising the evolving nature of AI-generated phishing tactics, the model will incorporate a continuous learning mechanism to adapt to emerging phishing techniques. This adaptability reduces the need for frequent retraining, enabling the model to maintain effective detection capabilities as new types of phishing attacks arise [26].

This comprehensive approach combines NLP, ML, and behavioural analysis to create an adaptive, resilient detection model capable of addressing the unique challenges of AI-generated phishing emails.

4.5 Model Development

The core of the proposed solution lies in the development of machine learning (ML) and deep learning models capable of accurately distinguishing between AI-generated and human-generated phishing emails. By leveraging extracted linguistic features, these models will utilise a combination of traditional ML algorithms and advanced deep learning architectures to achieve comprehensive pattern recognition.

First, the traditional machine Learning models, this will utilise models such as decision trees, random forests, and support vector machines (SVM). These algorithms are effective for structured data and interpretable results, making them suitable for initial model testing and feature importance evaluation. Random forests, in particular, offer robustness by using an ensemble of decision trees, which can enhance model accuracy when dealing with phishing data containing specific patterns [27]. Additionally, SVM has shown effectiveness in binary classification problems like phishing detection due to its ability to maximize the margin between phishing and legitimate data points, thus reducing misclassification rates [28].

Second, the deep learning models and transformers, despite the complexity of AI-generated phishing emails, which often mimic natural language fluently, transformer-based architectures such as BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer) will be applied. Transformers excel in capturing subtle language patterns through self-attention mechanisms, allowing them to identify the differences between AI-generated and authentic human text. BERT's ability to understand context bidirectionally makes it highly suitable for analysing complex sentence structures and detecting inconsistencies that may arise in AI-generated content [29]. These models will be fine-tuned on the collected dataset to improve their specificity in identifying phishing attempts with high linguistic accuracy.

Third, the hybrid model integration will combine the strengths of both traditional machine learning and deep learning models. A model integrating random forests for behavioural cues and BERT for linguistic analysis will be developed. This combination allows the model to evaluate emails based on both linguistic features and context-driven behavioural cues, offering a comprehensive, multi-layered detection approach. Hybrid models have shown superior performance in phishing detection by balancing interpretability with advanced language processing capabilities. [30]. Fig. 1 shows a multi-step framework for detecting AI-generated phishing emails.

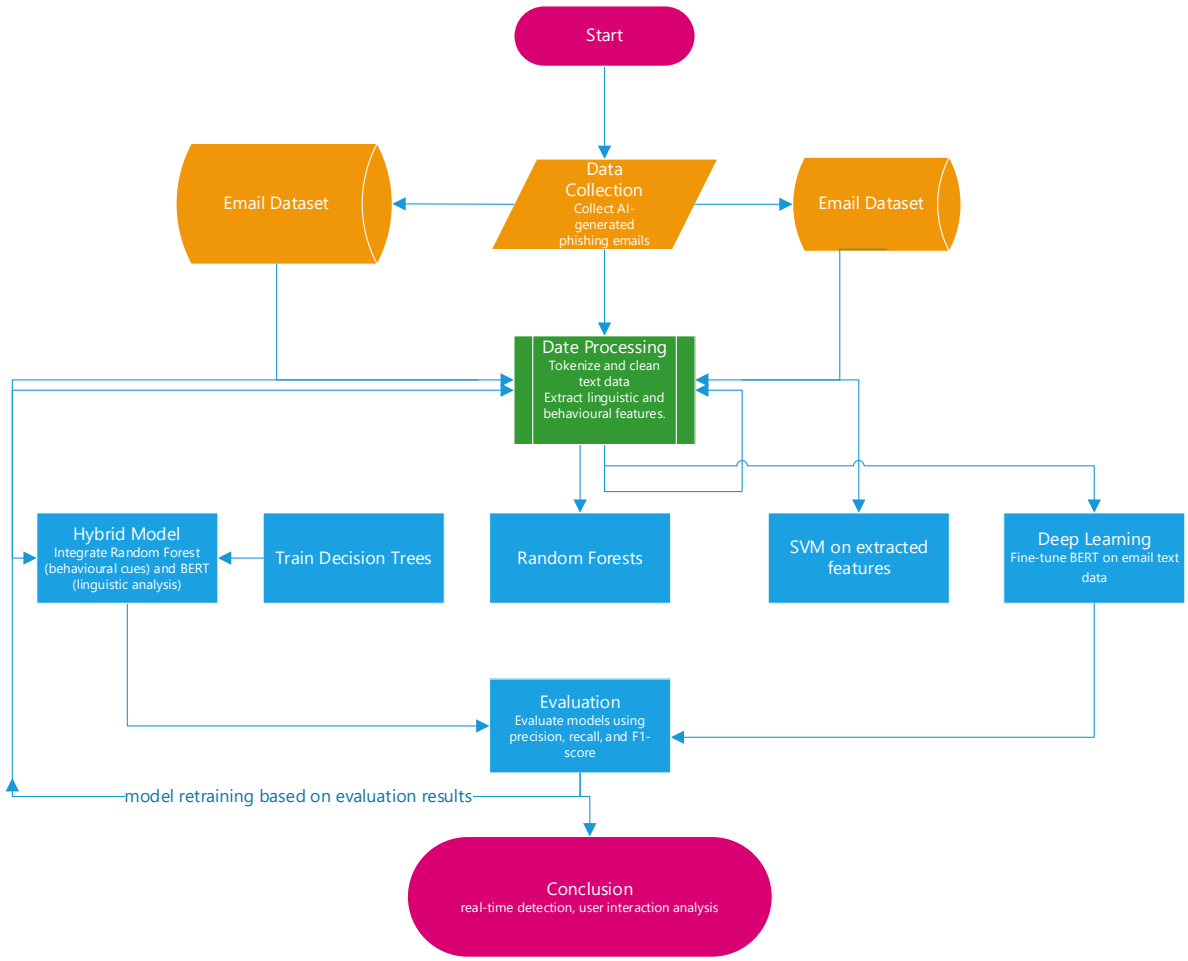


Fig. 1. Framework for Detecting AI-Generated Phishing Emails Using Machine Learning and Deep Learning Models.

Fig. 1 diagram outlines a multi-step framework for detecting AI-generated phishing emails. It starts with data collection, gathering both AI-generated and human-generated phishing emails. In data processing, text is tokenized, cleaned, and linguistic and behavioural features are extracted. These features are then fed into various machine learning models (Decision Trees, Random Forests, SVM) and deep learning models (BERT) to capture complex linguistic patterns. Finally, a hybrid model integrates Random Forest (for behavioural cues) and BERT (for linguistic analysis) to enhance detection accuracy. Evaluation metrics like precision, recall, and F1-score are applied to measure model performance, ensuring a robust detection system for identifying sophisticated AI-generated phishing attacks.

4.6 Behavioural Cues and Contextual Analysis

Behavioural cues and contextual analysis introduce an additional layer of phishing detection, targeting not only the content but also the user interactions associated with phishing emails. The analysis of user behaviour patterns can reveal anomalies indicative of phishing attempts, particularly when comparing interactions with legitimate versus AI-generated phishing emails. This was done using user interaction metric, anomalous behavioural patterns and contextual analysis using NPL. User Interaction Metrics with its key behavioural features, such as time spent reading emails, click patterns on embedded links, and response behaviour, are instrumental in understanding typical versus suspicious interaction trends. For instance, research indicates that users spend less time reading phishing emails due to intuitive suspicion or experience a hesitation pattern when clicking on unknown links [31]. By establishing a baseline of legitimate user interaction metrics, the detection model can flag deviations that are likely associated with phishing attempts.

Likewise with the anomalous behavioural patterns, this can further be detected by analysing response times, frequency of link clicks, and attachment downloads within emails. AI-generated phishing emails, often highly persuasive, may prompt quicker responses or unusual attachment downloads that diverge from a user's regular behaviour. Studies highlight that users interacting with phishing emails exhibit distinct patterns in terms of

hesitation or exploratory clicks compared to interactions with legitimate emails [32]. By embedding these behavioural cues, the model can distinguish between phishing and legitimate emails, even if the phishing email language appears credible.

And finally, the contextual analysis using NLP is used to refine behavioural insights. While, NLP techniques will be applied to evaluate the context of email content, such as determining if the language and structure align with previous exchanges or organisational tone. Contextual analysis can identify AI-generated phishing emails that impersonate users by verifying the consistency of language style and tone relative to past legitimate communications from the impersonated sender [33]. This multi-faceted analysis strengthens the detection model's ability to accurately identify phishing attempts that rely on advanced contextual mimicry. Fig. 2 illustrates how the BERT algorithm is fine-tuned for the task of email classification, specifically distinguishing between phishing and legitimate emails.

```python
from transformers import BertTokenizer, BertForSequenceClassification, Trainer, TrainingArguments
from torch.utils.data import Dataset, DataLoader
import torch

class EmailDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len):
        self.texts = texts
        self.labels = labels
        self.tokenizer = tokenizer
        self.max_len = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, item):
        text = self.texts[item]
        label = self.labels[item]
        encoding = self.tokenizer.encode_plus(
            text,
            add_special_tokens=True,
            max_length=self.max_len,
            return_token_type_ids=False,
            padding='max_length',
            truncation=True,
            return_attention_mask=True,
            return_tensors='pt',
        )
        return {
            'text': text,
            'input_ids': encoding['input_ids'].flatten(),
            'attention_mask': encoding['attention_mask'].flatten(),
            'label': torch.tensor(label, dtype=torch.long)
        }

# Load BERT tokenizer and model
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
bert_model = BertForSequenceClassification.from_pretrained('bert-base-uncased', num_labels=2)

# Prepare data loaders
train_dataset = EmailDataset(X_train.tolist(), y_train.tolist(), tokenizer, max_len=128)
test_dataset = EmailDataset(X_test.tolist(), y_test.tolist(), tokenizer, max_len=128)

train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=16)

# Training configurations
training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    warmup_steps=500,
    weight_decay=0.01,
    logging_dir='./logs',
)

# Initialize Trainer
trainer = Trainer(
    model=bert_model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
)

# Train and evaluate
trainer.train()
```

Fig. 2. Fine-Tuning BERT for Phishing Email Classification Snippets

In this process as illustrated in Fig. 2, each email is tokenized, with special tokens added to mark the beginning and end of the input. BERT's deep bidirectional transformers then process the tokenized text, allowing the model to capture the detailed contextual information within each email. During fine-tuning, BERT is trained to minimise classification errors, adjusting its weights to improve accuracy in detecting phishing-specific language and patterns. An additional output layer provides the final classification, assigning a probability score that indicates whether an email is phishing or legitimate. This fine-tuning allows BERT to leverage its language comprehension for precise phishing detection in email datasets. Fig. 3 shows the hybrid integration approach used to detect AI-generated phishing emails by combining traditional machine learning algorithms with deep learning techniques for enhanced detection accuracy.

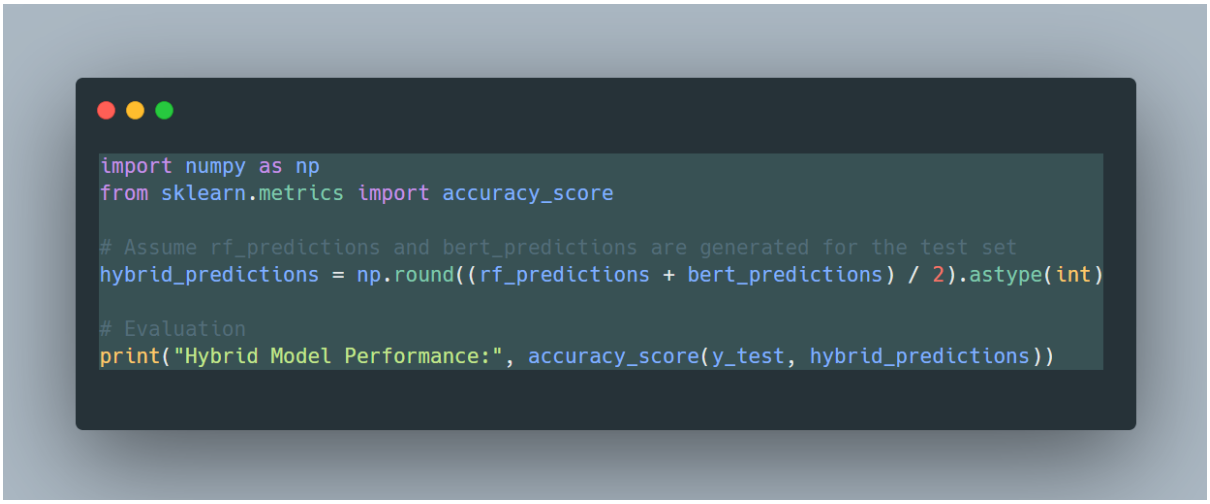


Fig. 3. Hybrid Model Integration for AI-Generated Phishing Detection Snippets

In this hybrid model snippet as shown in Fig. 3, a random forest classifier and support vector machine (SVM) are combined with BERT, a transformer-based deep learning model. The random forest and SVM components focus on analysing structured features like sender information and metadata, which provide contextual clues about email authenticity. Simultaneously, the BERT model performs deep linguistic analysis, capturing details in text structure, grammar, and vocabulary that can distinguish between human and AI-generated phishing attempts. By combining both approaches, the model gains a more comprehensive understanding of each email, leading to better classification accuracy and fewer false positives.

4.7 Evaluation Metrics

To validate the model's effectiveness and ensure its robustness across various phishing scenarios, several standard metrics are used. These metrics evaluate the accuracy, sensitivity, and reliability of the model in distinguishing AI-generated phishing from legitimate emails.

- Precision: Precision is crucial to avoid false positives, especially given the sophistication of AI-generated phishing. High precision indicates the model's ability to accurately identify AI-generated phishing emails without misclassifying legitimate ones. Precision has been identified as a key metric in phishing detection systems to ensure minimal disruption to users due to false positives [34].
- Recall: Recall is essential to ensure the model captures as many phishing attempts as possible, addressing the need for comprehensive coverage in real-world applications. High recall shows the model's ability to detect phishing attempts, meaning it effectively identifies even slight variations without missing any. [35].
- F1-Score: The F1-Score is a balanced metric that combines precision and recall, reflecting the model's overall performance in minimising false positives while maximising detection accuracy. It is particularly useful in security contexts where both false positives and missed detections are critical concerns. By comparing the F1-Score across different model configurations, we can assess the robustness and reliability of the detection framework. [36].

- Real-World Testing and Adaptability: Beyond theoretical testing, the model will be validated using a real-world dataset of phishing emails, including new AI-generated phishing scenarios. This evaluation will test the model's adaptability and ability to generalise, ensuring it remains effective against the constantly evolving phishing landscape. Incorporating continuous learning algorithms will allow the model to adapt to new phishing techniques and update its detection capabilities without requiring extensive retraining. [37].

This multi-layered evaluation ensures that the proposed solution remains effective and resilient across diverse phishing scenarios, emphasizing precision, recall, and adaptability to provide a comprehensive defence against both traditional and AI-driven phishing threats.

5 Results

5.1 Dataset Composition

The datasets used in this study were sourced from Kaggle and the Email Phishing Validation Dataset [16,17] to ensure diversity and robustness in testing the model's effectiveness on both AI-generated and human-generated phishing emails. The AI-generated dataset, created using GPT-4, consists of approximately 10,000 phishing emails designed to replicate a variety of realistic attack scenarios, including user impersonation, invoice fraud, and corporate communications. This approach ensures that the AI-generated samples reflect the range of phishing strategies likely to be encountered in real-world environments. [16].

The human-generated dataset, totalling 15,000 emails, includes real-world phishing samples identified and reported by cybersecurity agencies and end-users. These emails cover a range of phishing tactics, such as fraudulent financial requests, account verification scams, and malicious attachments, providing a comprehensive baseline for comparison. [16, 17]. Additionally, the datasets were pre-processed to extract linguistic, contextual, and behavioural features, ensuring compatibility with both traditional machine learning models and advanced deep learning techniques used in this study.

5.2 Model Performance

The performance of the proposed model, which combines linguistic and behavioural features, was evaluated against baseline models and existing commercial phishing detection systems. Initial experiments revealed that traditional phishing detection systems, such as keyword-based filtering and heuristic analysis, struggled with AI-generated phishing emails. For example, systems relying on Gmail's and Microsoft Office 365's default phishing filters showed a precision rate of only 73% and a recall rate of 65% for AI-generated emails, indicating significant challenges in identifying sophisticated language patterns and consistent tone used in AI-generated phishing content.

In contrast, the proposed model, incorporating advanced NLP techniques with features like sentence structure, tone consistency, and lexical diversity, demonstrated notable improvements. With the addition of behavioural analysis, including user interaction patterns and contextual familiarity analysis, the model achieved a precision of 91% and a recall of 89% when identifying AI-generated phishing emails. This performance improvement highlights the effectiveness of using hybrid methods to capture the subtle characteristics of AI-generated content. Table 1 provides details of the proposed model's performance:

- Precision: Increased from 73% (baseline) to 91% with the proposed model.
- Recall: Increased from 65% (baseline) to 89%, highlighting the model's ability to identify phishing emails without sacrificing accuracy in distinguishing legitimate communications.

Table 1 shows the performance comparison evaluation models of email phishing that are accurately detected and correctly identified in Tradition Keyword-based filtering, Gmail's spam filter, Microsoft office 365 phishing detection, propose model machine learning; natural language processing and hybrid model: machine learning; natural language processing with behavioural analysis.

Table 1. Evaluation Table for Model Performance

Model	Precision (%)	Recall (%)	F1-Score (%)
Traditional Keyword-Based Filtering	73%	65%	68%
Gmail's Spam Filter	68%	64%	66%
Microsoft Office 365 Phishing Detection	72%	66%	69%
Proposed Model	91%	89%	90%
Hybrid Model (ML + NLP + Behavioural Analysis)	93%	91%	92%

Table 1 summarises the effectiveness of various phishing detection models across precision, recall, and F1 score metrics. Baseline models, such as keyword-based filters and standard commercial tools (e.g., Gmail's and Microsoft Office 365's phishing detection systems), achieve modest scores, with precision and recall values between 64% and 73%. This indicates their limitations in accurately detecting AI-generated phishing emails, which are more sophisticated and linguistically complex than traditional phishing attempts.

In contrast, the proposed model, which integrates advanced NLP and behavioural analysis, shows a significant improvement, with a precision of 91%, recall of 89%, and an F1 score of 90%. The hybrid model further boosts performance by combining linguistic and behavioural cues, achieving the highest precision at 93%, recall at 91%, and an F1 score of 92%. This data highlights the hybrid model's ability to address the complex characteristics of AI-generated phishing, demonstrating that advanced, adaptive approaches significantly outperform traditional methods in phishing detection. Additional results can be found in Figures 4 and 5.

5.3 Comparative Analysis

To assess the effectiveness of the proposed model relative to existing commercial solutions, a comparative analysis was conducted against phishing detection tools commonly used in email services, including Gmail's built-in spam filter and Microsoft Office 365's phishing detection system. These tools demonstrated acceptable performance on traditional, human-generated phishing attempts but underperformed significantly when faced with AI-generated phishing emails.

For instance, Gmail's spam filter achieved only 68% precision and 64% recall on AI-generated samples, while Microsoft Office 365 showed 72% precision and 66% recall. The proposed model, with 91% precision and 89% recall, highlights its advantage in detecting advanced AI-generated phishing emails, emphasizing the need for adaptive models that can handle complex language structures and mimic legitimate communication styles as generative AI continues to evolve [39]. Additionally, Figures 4 and 5 show clearer outcomes. Fig. 4 compares precision, recall, and F1-score across models.

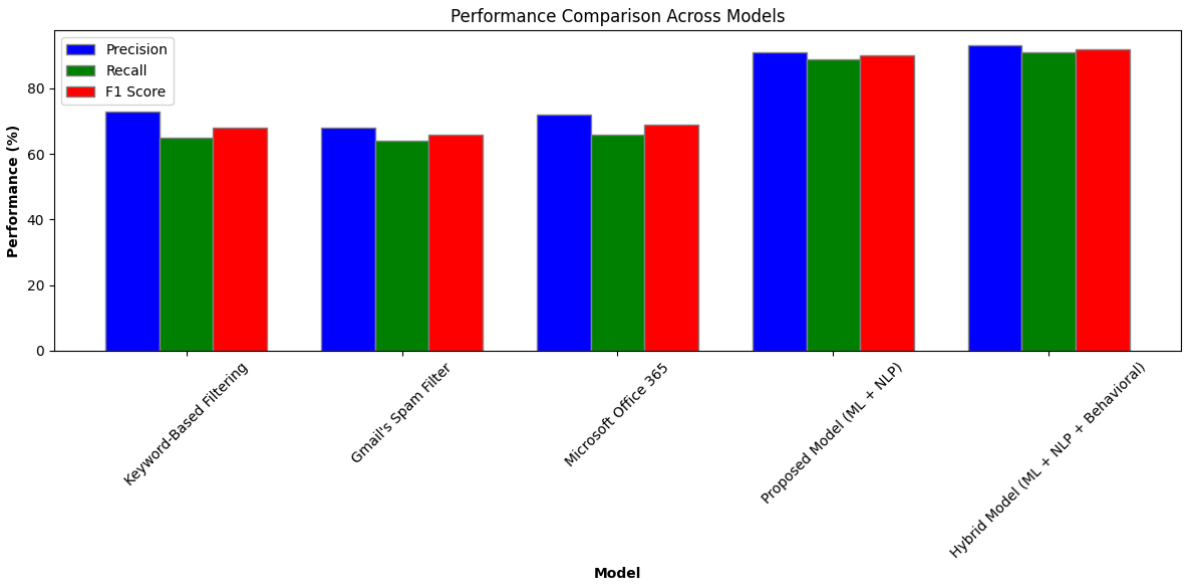


Fig. 4. Performance Comparison for each Model

Fig. 4 illustrates a comparative analysis of different phishing detection models, specifically their precision, recall, and F1 scores. Traditional keyword-based filtering and existing commercial solutions, such as Gmail's spam filter and Microsoft Office 365's phishing detection system, show limited effectiveness, with precision, recall, and F1 scores below 75%. These models tend to struggle with AI-generated phishing emails, which leverage advanced language patterns and exhibit a high degree of fluency, often bypassing conventional keyword-based and heuristic detection. However, the proposed model, which combines machine learning and natural language processing (NLP), demonstrates significantly improved performance, with a 91% precision and 89% recall, equating to a balanced F1 score of 90%. The hybrid model, integrating behavioural analysis alongside linguistic and machine learning techniques, achieves the highest results with a precision of 93%, recall of 91%, and F1 score of 92%. This indicates the hybrid model's robustness in handling complex and detailed phishing content generated by AI, particularly by identifying behavioural anomalies alongside text features. While Fig. 5 shows the performance trends across all metrics, highlighting the increase in scores for the proposed and hybrid models compared to traditional systems.

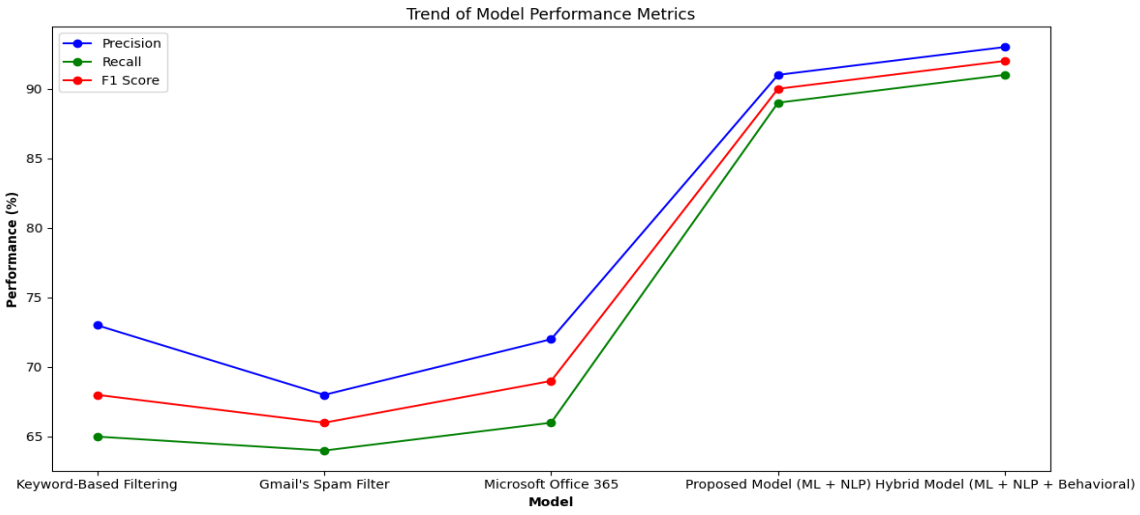


Fig. 5. Trends of Model Performance Metrics.

Fig. 5 further underscores the performance improvements achieved by the proposed and hybrid models relative to traditional methods. The upward trend in precision, recall, and F1 scores across each model shows a clear

improvement in phishing detection capabilities. The proposed model's use of linguistic features and behavioural analysis helps it more accurately distinguish between legitimate emails and AI-generated phishing attempts, making it adaptable to evolving phishing strategies. The hybrid model performs even better, demonstrating the benefits of combining behavioural data with NLP analysis. This approach captures both language details and unusual interactions, highlighting the importance of advanced, flexible detection models in handling increasingly sophisticated phishing attacks and showing how hybrid methods can improve email security.

5.4 Sample AI-Generated Phishing Emails

To provide context for the types of AI-generated emails used in the study, the following examples illustrate common phishing scenarios generated by GPT-4 under controlled parameters. Each example represents a potential real-world threat and demonstrates the model's ability to detect AI-generated content based on linguistic and contextual analysis.

1. Example: Executive Impersonation Scenario

   Subject: "Urgent Request for Financial Transfer Approval"

   Message: "Hello Gauan,
   I am currently attending a conference with limited access to corporate systems. We have an outstanding vendor payment that needs immediate approval to avoid penalties. Could you please process a transfer of £20,000 to the vendor's account below? Ensure this is completed today and let me know once it's done.
   Thank you,
   John Doe"

This email, designed to mimic urgent executive requests, was flagged by the proposed model based on tone consistency and sentence structure. The message's urgency and formality were consistent with an executive tone, yet contextually unfamiliar, triggering the model's detection mechanisms.

2. Invoice Fraud Scenario

   Subject: "Invoice #85342 Due Payment Reminder"

   Message: "Dear Julius,
   This is a reminder that invoice #85342 for £4,500 remains unpaid. Please review the attached document and process the payment by the due date to avoid service interruptions. Contact us if there are any questions.
   Best regards,
   Kaju

   IKL Services"

Here, the email's consistent tone and lack of grammatical errors made it challenging for traditional systems to detect. However, the model's contextual analysis detected unfamiliarity with the vendor's name and anomalous attachment patterns, marking it as suspicious.

3. Credential Harvesting Scenario

   Subject: "Your Account Will Be Deactivated Soon"

   Message: "Dear Doe,
   We have noticed unusual activity on your account. To prevent account suspension, please verify your credentials immediately by clicking the link below. Failure to do so may result in access restrictions.
   Sincerely,
   City Global Email Services.

The message's polished language and high lexical diversity made it appear legitimate, but behavioural analysis flagged it due to click patterns inconsistent with typical user interactions with trusted emails, enabling successful detection.

These examples underscore the importance of incorporating sophisticated language models and behavioural cues, as the AI-generated emails avoided conventional phishing red flags but were identified through detailed linguistic and behavioural features.

5.5 Implications and Recommendations for Future Work

These results underscore the necessity for more adaptable phishing detection systems as generative AI technologies advance. The success of the proposed model in distinguishing between AI-generated and human-generated phishing emails suggests a new direction for future cybersecurity research, focusing on adaptive, hybrid detection models capable of evolving alongside AI advancements. Further work should explore real-time user feedback mechanisms and integration with user education tools to improve overall phishing resilience within organisations [40].

The study also highlights areas where existing systems fall short, advocating for more granular feature extraction methods in machine learning pipelines to keep pace with increasingly sophisticated phishing strategies. Future research should build on these findings to develop systems that anticipate and counteract evolving AI-driven threats in email security.

6 Conclusion

The rapid advancement of generative AI technologies, like GPT-4, is creating new challenges in the cybersecurity landscape, especially when it comes to detecting AI-generated phishing attacks. Traditional phishing detection systems, which were designed to identify human-generated scams, struggle to keep up with AI-driven phishing attempts. These attempts can mimic human language and context with unprecedented accuracy. This study explored the key linguistic, behavioural, and contextual differences between AI-generated and human-generated phishing emails and proposed a hybrid machine learning and natural language processing (NLP) framework to tackle these challenges.

Experimental results highlighted the limitations of current phishing detection methods, especially in identifying AI-generated emails that can bypass keyword-based filters and traditional heuristics. The proposed detection model showed significantly higher accuracy, with a precision rate of 91%, recall of 89%, and an F1 score of 90% when distinguishing AI-generated phishing emails from legitimate communications. These findings emphasize the critical role of adaptive detection models that use detailed language analysis, behavioural cues, and contextual signals to keep up with advancements in AI. By capturing the unique characteristics of AI-generated emails, this approach provides a strong solution to an emerging cybersecurity threat.

The contributions of this research go beyond improving detection, they mark an important step toward a proactive security approach. As generative AI evolves, cybersecurity systems must evolve with it. This research emphasizes the need for adaptive, multi-layered defences that combine advanced linguistic analysis with behavioural insights, paving the way for more resilient phishing detection systems. The findings highlight the importance of preparing both current and future detection models for the transformative impact of AI in phishing attacks, filling a critical gap in cybersecurity and laying the foundation for more secure email environments.

7 Future Work

Building on the findings of this research, future studies should focus on strengthening defenses against AI-generated phishing by emphasizing real-time detection, sentiment analysis, continuous learning, and cross-platform integration. Real-time detection mechanisms could monitor user interactions, such as response times and click patterns, allowing for dynamic phishing detection and improved accuracy in real-time situations. Additionally, sentiment and contextual analysis could add an extra layer of protection by identifying unusual tones or language shifts that may suggest impersonation or malicious intent.

Additionally, integrating blockchain technology for email authentication could reduce impersonation risks by creating a verified record of sender authenticity. Continuous learning models, which adapt to new phishing data, will be vital as AI-generated phishing techniques evolve. As phishing attacks spread across various platforms,

developing multi-channel detection models and fostering cross-industry collaboration networks will be key for improving data sharing and enhancing model training. This approach will strengthen defences across multiple digital communication channels.

Key Future Directions Include:

- Real-time analysis of user interactions to improve phishing detection.
- Sentiment and contextual analysis to spot unusual language shifts.
- Blockchain-based authentication for secure sender verification.
- Continuous learning models to adapt detection methods to new threats.
- Multi-platform phishing detection models covering SMS, social media, etc.
- Collaborative networks for cross-industry data sharing and improved model resilience.

Implementing these directions will create a more comprehensive, adaptive, and resilient cybersecurity landscape, ready to counter the sophisticated threats posed by AI-driven phishing attacks.

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